

Drugs Used in Disorders of Coagulation

Hemostasis is the process of **forming clots** in the walls of **damaged blood vessels** and preventing **blood loss** while maintaining blood in the **fluid state** within the vascular system

two main components of hemostasis.

Primary hemostasis

- **platelet aggregation and platelet plug formation.**

Secondary hemostasis

- **deposition of insoluble fibrin, which is generated by the proteolytic coagulation cascade**

Hemostasis...

Evenest in hemostasis

- Vascular constriction
- Formation of platelet plug
- Formation of blood clot
- Growth of fibrous tissue into the clot
- **Fibrin degradation**

Vascular constriction

❖ The contraction results from:-

- Local myogenic spasms
- Local autacoid factors
- Nervous reflexes

❖ Platelets release, *thromboxane-A2* which is responsible for vasoconstriction of smaller vessels

❖ Mechanism of platelet plug

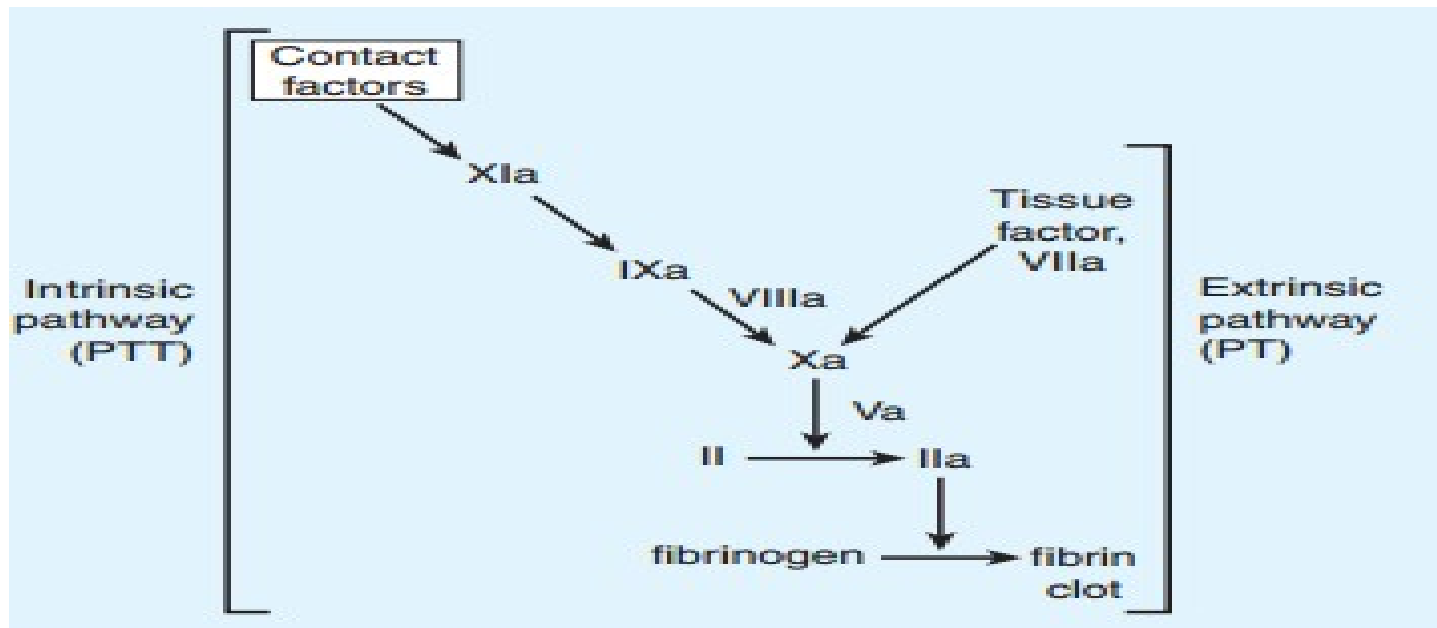
- Platelet adhesion
- Platelet activation
- Platelet aggregation
- Formation of temporary haemostatic plug

Blood coagulation

- ❖ Blood coagulates is the transformation of soluble fibrinogen into **insoluble** fibrin by the enzyme **thrombin**.
- ❖ The clot begins to develop
 - in **15 to 20** seconds if the trauma to the vascular wall has been severe, and in
 - **1 to 2** minutes if the trauma has been **minor**.
- ❖ Two pathways **Intrinsic pathway & Extrinsic pathway**

Initiation of Clotting

- ❖ The main initiator of blood coagulation in vivo is the tissue factor (TF)-factor VIIa pathway
- ❖ Intrinsic path way and extrinsic pathway converges at factor X



Clotting in Vivo

Clotting factors

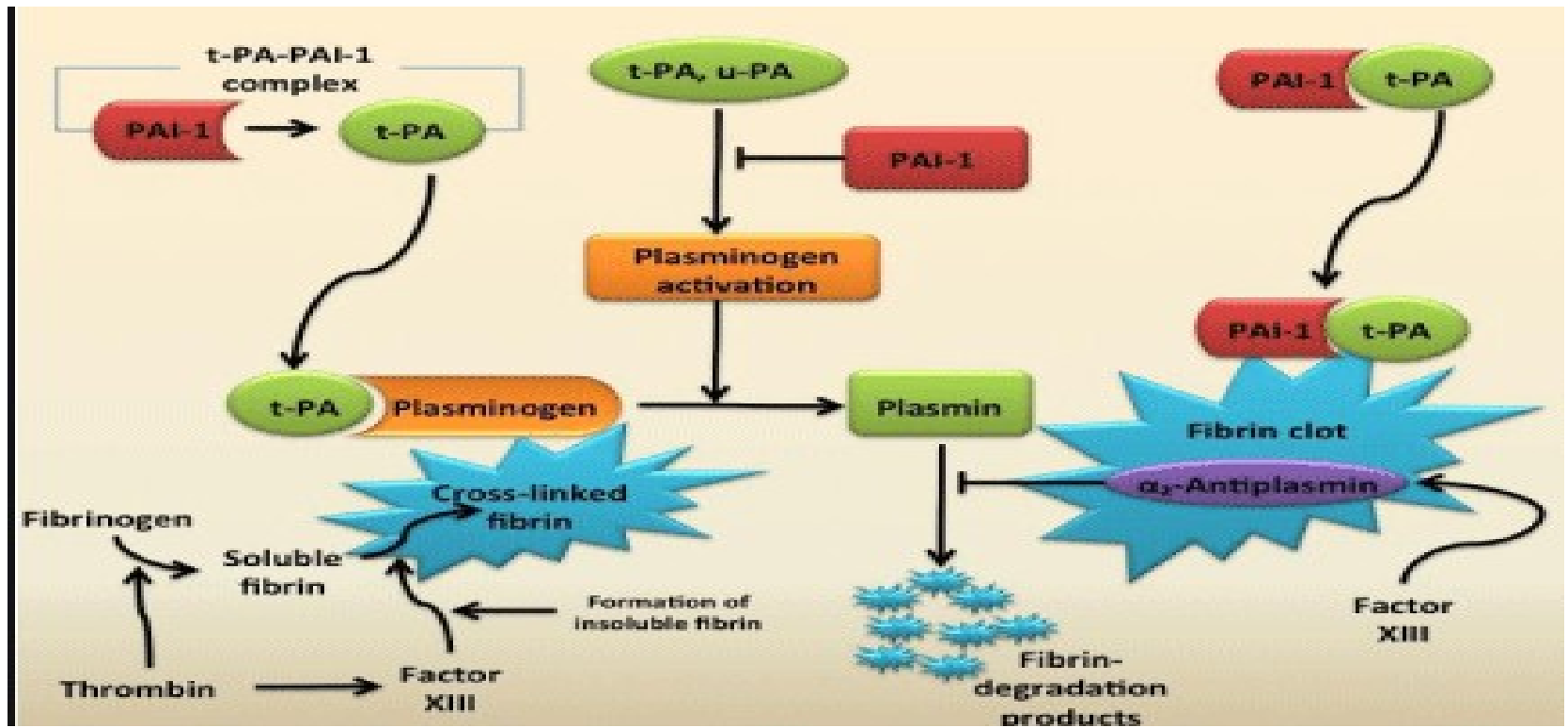
Component or Factor	Common Synonym	Target for the Action of:
I	Fibrinogen	
II	Prothrombin	Heparin (IIa); warfarin (synthesis)
III	Tissue thromboplastin	
IV	Calcium	
V	Proaccelerin	
VII	Proconvertin	Warfarin (synthesis)
VIII	Antihemophilic factor (AHF)	
IX	Christmas factor, plasma thromboplastin component (PTC)	Warfarin (synthesis)
X	Stuart-Prower factor	Heparin (Xa); warfarin (synthesis)
XI	Plasma thromboplastin antecedent (PTA)	
XII	Hageman factor	
XIII	Fibrin-stabilizing factor	
Proteins C and S		Warfarin (synthesis)
Plasminogen		Thrombolytic enzymes, amino-caproic acid

Fibrinolysis

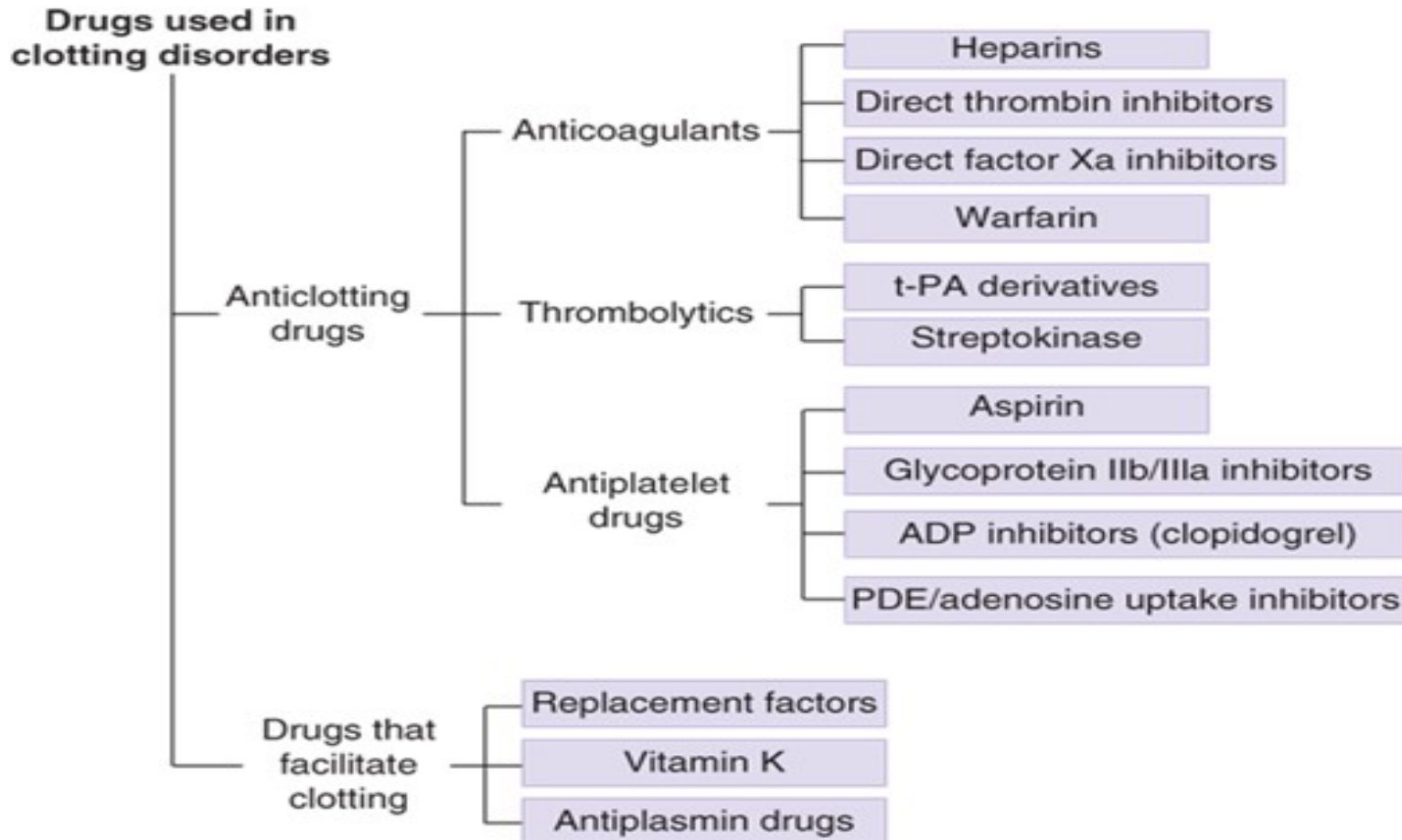
- ❖ **Fibrinolysis** refers to the process of fibrin digestion by the fibrin specific protease, plasmin.
- ❖ Occur after bleeding is ceased with fibrin formation
- ❖ **Tissue plasminogen activator, urokinase, and streptokinase** activate the fibrinolytic system
- ❖ **α 2 antiplasmin** circulates in the blood at high **concentrations** and under physiologic conditions
 - Block/inhibit fibrinolysis

Fibrinolysis...

Fig: fibrinolytic system



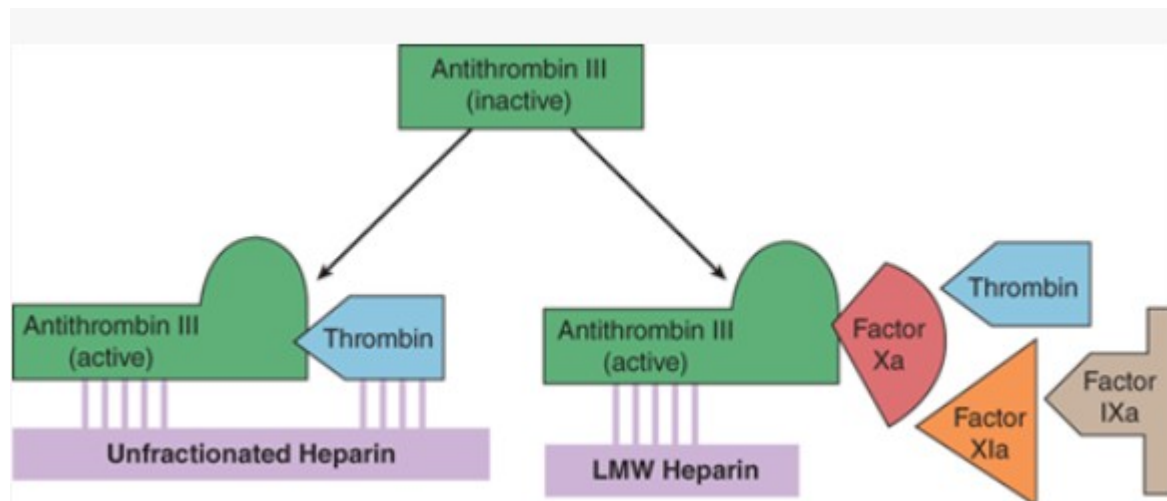
Drugs used in Coagulation Blood disorder

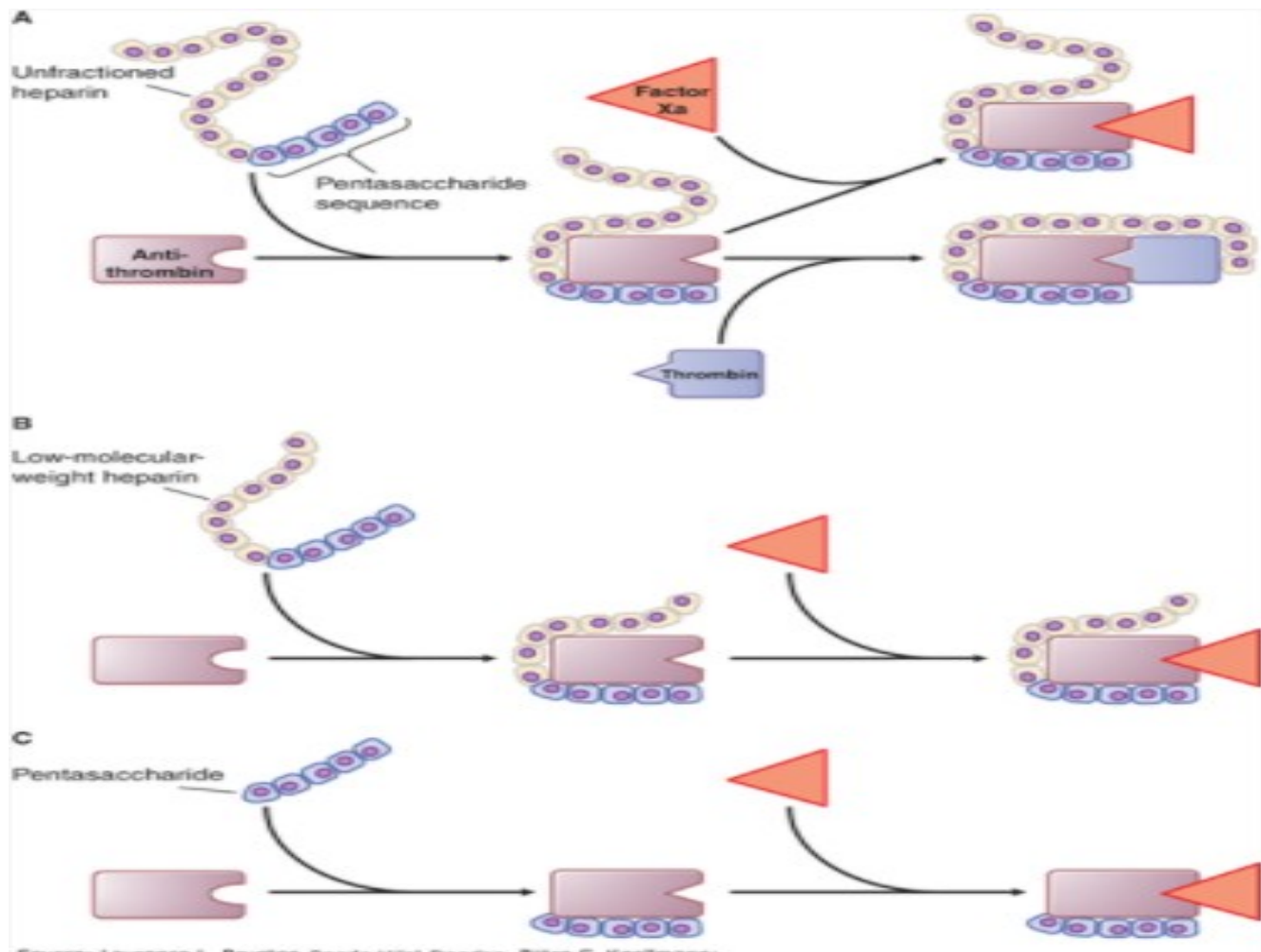


Heparin

- ❖ Heparin and related drugs are.... indirect thrombin inhibitors
 - ✓ their antithrombotic effect is exerted by their interaction with a separate protein, [antithrombin](#).
 - ✓ All bind to antithrombin III
- ❖ Classifications
 - ✓ Unfractionated heparin (UFH),
 - also known as high-molecular-weight (HMW) heparin,
 - ✓ low-molecular-weight (LMW) heparin, and
- Example of LMW..... enoxaparin, dalteparin, and tinzaparin
- ✓ the synthetic pentasaccharide fondaparinux
- ❖ fondaparinux bind to antithrombin and enhance its inactivation of factor Xa

- ❖ Activated antithrombin III (AT III) degrades thrombin, factor X, and several other
- ❖ Binding of these drugs to AT III can increase the catalytic action of AT III 1000-fold





Miscellaneous Pharmacological Effects

- ❖ High doses of heparin can interfere with platelet aggregation and prolong the bleeding time.
- ❖ In contrast, LMWH and fondaparinux have little effect on platelets.

Chemistry & Mechanism of Action of Heparin

- ❖ Its biologic activity is dependent upon the endogenous anticoagulant antithrombin
- ❖ Antithrombin inhibits the following clotting factor
thrombin (IIa), IXa, and Xa
- In the absence of heparin, these reactions are slow; in the presence of heparin, they are accelerated **1000-fold**

Heparin..

- ❖ In comparison with UFH—LMWHs
 - ✓ have equal efficacy,
 - ✓ increased bioavailability from the subcutaneous site of injection, and
 - ✓ less frequent dosing requirements (once or twice daily is sufficient).

Heparins...

Clinical Pharmacology

❖ Heparin is indicated

✓ for prevention and treatment of **venous thromboembolic disease**, arterial thrombosis, and prevention of **thrombosis** in arterial or cardiac surgery.

PK

- Continuous IV infusion or sc injection
- The clearance involves
 - **A rapid, saturable** : binds to plasma proteins, endothelial cells & macrophages
 - **A slower, unsaturable** : Renal
- $t_{1/2}$: $\uparrow$ with $\uparrow$ of the dose

Monitoring of UFH Therapy

- **Initiation phase:**

- Should be ordered **every 6 hrs** until the result falls within the **target therapeutic range**
- Whenever the dosage is changed, the **aPTT** should be reevaluated **every 6 hrs** until the desired therapeutic range is reached

- **Stable phase:**

- The test should be ordered **once a day** until UFH therapy is discontinued

Toxicity of Heparin

Bleeding and miscellaneous effects

- ❖ The major adverse effect of heparin is bleeding.
 - ✓ More in elderly women and patients with renal failure
- ❖ allergy.
- ❖ Increased **loss of hair and reversible alopecia**
- ❖ Long-term heparin therapy is associated with osteoporosis and spontaneous fractures
- ❖ long-term use is associated with mineralocorticoid deficiency.

Toxicity of Heparin

Heparin-induced thrombocytopenia (HIT)

- ❖ Occurs In 1–4% of individuals treated with UFH. Surgical patients are at greatest risk
- ❖ Morbidity and mortality in HIT...due to venous
 - If an **indwelling catheter** is present, the risk of thrombosis is increased in that extremity.
- ❖ **HIT is higher in**
 - ✓ UFH than LMWHs
 - ✓ in individuals treated with **bovine origin of** UFH of than **porcine** heparin
- ❖ **Patients who develop HIT are treated by**
 - ✓ discontinuance of heparin and administration of the direct thrombin inhibitor **argatroban**.

Heparin is contraindicated in

- ✓ patients with HIT,
- ✓ hypersensitivity to the drug,
- ✓ active bleeding, hemophilia
- ✓ purpura
- ✓ ulcerative lesions of the GIT,
- ✓ visceral carcinoma
- ✓ advanced hepatic renal disease
- ✓ intracranial hemorrhage,
- ✓ infective endocarditis,

Reversal of Heparin Action

- ❖ Excessive anticoagulant action of heparin is treated by discontinuance of the drug.
- ❖ If bleeding occurs,administer protamine sulfate
 - ✓ For LMWHs....**Protamine sulfate** is ineffective
 - ✓ Protamine will not reverse the action of **fondaparinux**
 - ✓ **danaparoid** can be removed by **plasmapheresis**.

Drugs in this class: Hirudin, bivalirudin, Argatroban ,melagatran, dabigatran (oral)

MOA: exert their anticoagulant effect by directly binding to the active site of thrombin

Parenteral direct thrombin inhibitors

Hirudin. irreversible [thrombin](#) inhibitor from leech saliva and now available in recombinant form as **lepirudin**

- ❖ Lepirudin has little effect on **platelets** or the bleeding time
- ❖ Approved for use in patients with thrombosis related to **heparin-induced thrombocytopenia (HIT)**

Lepirudin...

- ❖ It is excreted by the **kidney** and should be used with great caution in patients with **renal insufficiency** as no antidote exists.
- ❖ **Up to 40%** of patients who receive long-term infusions develop an antibody directed against the **thrombin-lepirudin** complex.
 - The complex not cleared from the body and can result in an enhanced **anticoagulant effect**
- ❖ Some patients **re-exposed** to the drug developed **life-threatening anaphylactic reactions**.

Bivalirudin

- ❖ It has **rapid onset** and offset of action.
- ❖ The drug has a short half-life with clearance that is **20% renal**
- ❖ Given IV
- ❖ It **inhibits platelet activation** and
- ❖ has been FDA-approved for use in percutaneous coronary angioplasty

Argatroban

- ❖ approved for use in patients with HIT with or without thrombosis and coronary angioplasty in patients with HIT.
- ❖ IT has a very **a short half-life.....**is given by **continuous IV**
- ❖ monitored by aPTT.
- ❖ Its clearance is **not affected by renal disease** but is dependent on **liver function**;
 - dose reduction is required in patients with **liver disease**.

Oral direct thrombin inhibitor

Drug: Dabigatran etexilate

- ❖ Advantages of oral direct [thrombin](#) inhibition include
 - ✓ Predictable pharmacokinetics and bioavailability,
 - ✓ Like direct parenteral thrombin inhibitor it has rapid onset and offset of action
- PK and PD
- ✓ The oral bioavailability.... 3–7%
- ✓ Half life...12-17 hrs
- ✓ Renal impairment results in prolonged drug clearance.
 - ❑ Dosage and administration is based on CrCl
 - CrCl > 30 mL/min.....150 mg BID
 - 15–30 mL/min.....75 mg BID
- ✓ Drug interaction....ketoconazole increases the half life of dabigatran and should be avoided in renal impairment
- ✓ Reversing the action of dabigatran.... **Idarucizumab**

- ✓ Dabigatran is approved for
 - ❖ reduction in risk of **stroke** and **systemic embolism** with **nonvalvular atrial fibrillation**,
 - ❖ treatment of VTE following **5–7 days** of initial [heparin](#) or LMWH therapy,
 - ❖ reduction of the risk of recurrent **VTE**, and **VTE** prophylaxis following hip or **knee replacement surgery**.

Dabigatran etexilate

- ❖ Efficacy is equivalent with warfarin therapy
- ❖ Has fewer drug interaction
- ❖ Compared to parenteral short half life newer anticoagulants, Dabigatran
 - ✓ Has better compliance (because it is oral agent)
 - ✓ Do not require monitoring

Vitamin K Antagonists: Warfarin

- **MOA:**
 - Block the γ -carboxylation of several glutamate residues in the synthesis of six coagulation-related proteins in the liver:
 - Coagulant factors: F-II (prothrombin), VII, IX & X
 - Anticoagulant regulatory proteins: proteins C & S
- It **does not** affect circulating functional clotting factors
 - Onset of action starts when these factors have been eliminated

Factor	II	VII	IX	X	Protein C
Half-life (hrs)	60	6	24	40	14

Warfarin....

- **PK**

- Have a slow onset & offset of action
- GI absorption is rapid & complete
- Volume of distribution is very small
- **99 % protein bound**
- Long $t_{1/2}$ in plasma (36 hrs)
- Duration of action: 2- 5 days
- Metabolized by the CYP450 enzymes

- **Major use**

- **Prevention & Tx of thromboembolic disease**

Warfarin Drug interaction

- ❖ The coumarin anticoagulants often interact with other drugs and with disease states.
- ❖ These interactions can be
 - pharmacokinetic and pharmacodynamic effects
- ❖ Pharmacokinetic
 - P450 CYP2C9 enzyme induction, enzyme inhibition, and reduced plasma protein binding.
- ❖ Pharmacodynamic mechanisms for interactions with warfarin are **synergism** (impaired hemostasis, reduced clotting factor synthesis, as in hepatic disease), **competitive antagonism (vitamin K)**, and **an altered physiologic control loop for vitamin K** (hereditary resistance to oral anticoagulants).

Warfarin Drug interaction

❖ The most serious interactionsthose that (+)risk of bleeding.

- interactions with the mostly pyrazolones phenylbutazone and sulfinpyrazone.
- **hypoprothrombinemia** and **inhibit platelet function** and may induce peptic ulcer disease

Mechanism of the above drug interaction

- Inhibition of oxidative metabolic transformation of S-warfarin
- **displacement** of albumin-bound warfarin...Increase free warfarin

Metronidazole, fluconazole, and trimethoprim-sulfamethoxazole

- inhibit the metabolic transformation of S-warfarin

amiodarone, disulfiram, and cimetidine.....inhibit metabolism of enantiomorphs of warfarin

Warfarin Drug interaction....

Aspirin.....increase the effect by its antiplatlet effect

Third generation cephalosporins also increase warfarin effect

Drug interaction those decrease warfarin effect

Barbiturates and rifampin

- induction of the hepatic enzymes

Cholestyramine

- Decrease intestinal absorption

PD drug interaction of Warfarin

❖ Pharmacodynamic reductions of anticoagulant effect occur with

- ✓ increased vitamin K intake (increased synthesis of clotting factors)
- ✓ chlorthalidone and spironolactone (clotting factor concentration),
- ✓ hereditary resistance (mutation of vitamin K reactivation cycle molecules)

PD increment of warfarin effect

- ✓ hepatic disease, and hyperthyroidism....increase effect by increasing the turnover rate of clotting factor

Monitoring of Warfarin Effect...

- ❖ PT screens the extrinsic pathway of the coagulation system
- ❖ It is performed by adding calcium & tissue thromboplastin to a sample of citrated, platelet-poor plasma & measuring the time required for fibrin clot formation
- ❖ **It is most sensitive** to deficiencies in the vitamin K-dependent clotting factors II, VII, IX, & X
 - It is also sensitive to deficiencies of factor V

Monitoring of Warfarin Effect...

- Efforts to standardize & report the PT as an INR (International Normalized Ratio) depend on assigning reagents an International Sensitivity Index (ISI) so that:

$$\text{INR} = [\text{PT patient} / \text{PT normal}]^{\text{ISI}}$$

- ❖ The recommended INR
- ❖ for prophylaxis and treatment of thrombotic disease is **2–3**.
- ❖ Patients with some types of artificial heart valves (other medical conditions increasing thrombotic risk have a recommended range of **2.5–3.5**

Monitoring of Warfarin Effect...

- During initiation of therapy:
 - INR should be checked every 2 -3 days for 1-2 wks until a stable therapeutic effect is achieved
- During chronic therapy:
 - INR should be monitored every 2-3 wks depending on stability of the response & minor dose adjustments are frequently needed

Adverse reactions & its management

- **Adverse Reactions: Most Common**

- Bleeding

- **Adverse Reactions: Rare/Severe/Important**

- Intracranial, retroperitoneal, or intraocular bleeding, purple toe syndrome, skin necrosis

- **Reversal of Warfarin Action**

- Stopping the drug & administering vitamin K, fresh-frozen plasma, prothrombin cpx concentrates & recombinant factor VIIa

Oral direct factor Xa inhibitors

Drugs in this class include.....rivaroxaban, apixaban, edoxaban, and betrixaban

Reversal of Anti-Xa Drug Effect.... **Andexanet alfa**

Alternative.....Four-factor concentrate

Rivaroxaban..... has high oral bioavailability with food

- ✓ the peak plasma level.....2–4 hours;
- ✓ has extensively protein-bound
- ✓ It is a substrate for the cytochrome P450 system and the P-glycoprotein transporter.
 - Drug that inhibit CYP3A4 and P-glycoprotein....increase effect of rivaroxaban....example Ketoconazole

Oral direct factor Xa inhibitors

Apixaban....

- ✓ oral bioavailability50% and prolonged
- ✓ half-life of 12 hours with chronic dosing.
- ✓ Same to rivaroxaban....drugs inhibiting both **CYP3A4** and **P-glycoprotein**...increase its effect

Edoxaban

- ✓ **It** is a once-daily Xa inhibitor
- ✓ Oral bioavailability...62%
- ✓ Peak drug concentrations.....1–2 hours after dosage and are not affected by food.
- ✓ The drug half-life.....10–14 hours.
- ✓ does not induce CYP450 enzymes.
- ✓ No dose reduction is required with concurrent use of **P-glycoprotein inhibitors**

Oral direct factor Xa inhibitors

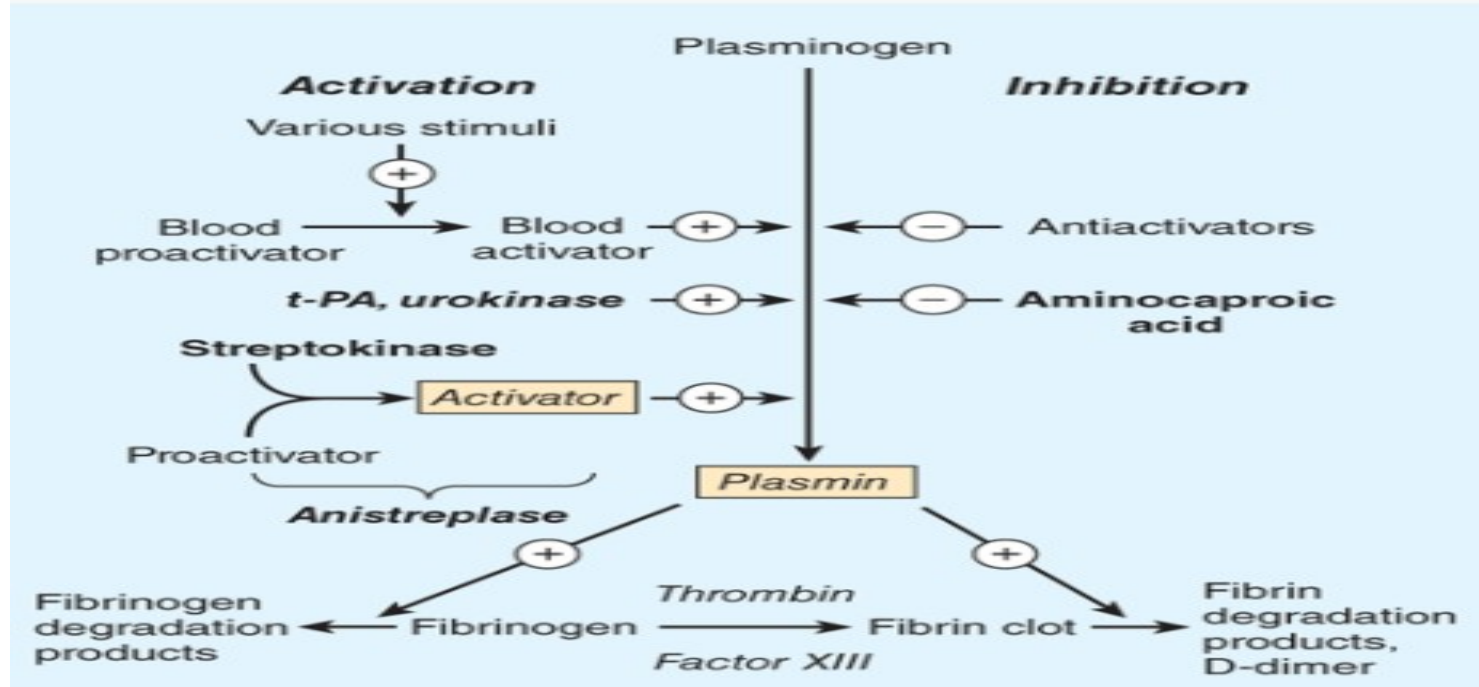
Betrixaban

- ✓ It is a once daily Xa inhibitor with
- ✓ **Food reduced its effect by 70 %**
- ✓ oral bioavailability....34%
- ✓ half-life of 19–27 hours;
- ✓ peak levels are achieved in 3–4 hours,
- ✓ Dose reduction is required if given with **P-glycoprotein inhibitors** and in **renal failure**.

Fibrinolytic (Thrombolytic) drugs

❖ Fibrinolytic drugs rapidly lyse thrombi by **catalyzing** the formation of the serine protease **plasmin** from its precursor zymogen, plasminogen

Drugs in this class: Streptokinase, Urokinase, alteplase. Reteplase



Thrombolytic Drugs..



- **Alteplase (tPA)**
 - Clot specific, acting mainly on fibrin-bound plasminogen,
 - no allergy problem
- **Clinical features**
 - Overriding factor in effectiveness is early administration e.g. >60% ↓ mortality post –MI if used within **3hrs**

Thrombolytic Drugs..

- **Sterptokinase**

- Acts on both bound & free plasminogen (not clot specific), depleting circulating plasminogen & factors V & VIII
- Is antigenic (foreign protein derived from β - hemolytic stereptococci)

Thrombolytic Drugs...

- **Clinical features...**

- Complications include bleeding, possible intracerebral hemorrhage
- Streptokinase may cause hypersensitivity reactions & hypotension
- Antifibrinolysin (**aminocaproic & tranexamic acids**)- possible antidotes in excess bleeding

Antiplatelet agents

❖ Classifications of Antiplatelet agents

✓ inhibition of prostaglandin synthesis

- Aspirin

✓ inhibition of ADP-induced platelet aggregation

- clopidogrel, prasugrel, ticlopidine

✓ blockade of glycoprotein IIb/IIIa (GP IIb/IIIa) receptors on platelets

- abciximab, tirofiban, and eptifibatide

Drugs used for bleeding disorder

Drugs /class of drugs:

Vitamin K, Plasma Factors, Desmopressin acetate, Cryoprecipitate, Fibrinolytic inhibitors

Vitamin-K

- ❖ Vitamin K is responsible for normal biological activity of Factor II, VII, IX and X
- ❖ It is a fat-soluble substance found primarily in leafy green vegetables.
- ❖ Exists in two forms
 - ✓ vitamins K1 (**phytonadione**)....found in food
 - ✓ and K2 (menaquinone)..... found in **human tissues** and is synthesized by **intestinal bacteria**.

Vitamin-k

- ❖ **Vitamins K1 and K2** require **bile salts** for absorption from the intestinal tract
- ❖ Vitamin K1 is available clinically in oral and parenteral forms
- ❖ Complete effect by vitamin k...after 24 hrs in reversing warfarin effect
- ❖ Both IV/PO can achieve Vitamin K repletion
 - ✓ SC administration is erratic
- ❖ IV vitamin K1 should be slow
 - ✓ rapid infusion can produce dyspnea, chest and back pain, and even death.

Fibrinolytic inhibitors

Drugs in this class: Aminocaproic acid, Tranexamic acid

Aminocaproic acid

- ✓ It is rapidly absorbed orally
- ✓ cleared from the body by the kidney
- ✓ oral dosage....6 g QID
- ✓ IV.....5-g loading dose should be infused over 30 minutes
 - to avoid hypotension.

Tranexamic acid

- ✓ It is an analog of aminocaproic acid

Dosage....orally with a **15-mg/kg** loading dose followed by 30 mg/kg every **6H**

Therapeutic uses Fibrinolytic inhibitors

- ✓ prophylaxis for rebleeding from intracranial aneurysms
- ✓ postsurgical GI bleeding
- ✓ postprostatectomy bleeding
- ✓ bladder hemorrhage

Adverse effects

- ✓ intravascular **thrombosis** from inhibition of plasminogen activator,
- ✓ hypotension, myopathy, abdominal discomfort, diarrhea, and nasal stuffiness.